



Management School

**The Variance Risk Premium and Implied Volatility Term Structure:
Regime-Conditional Dynamics and Hump Effect**

This dissertation is submitted in part fulfilment of the

MSc Finance and Investment Management

I clarify that this dissertation is entirely my own work

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Abstract

In this study, we examine whether the shape of the S&P 500 Implied volatility term structure can predict the returns earned by investors who bear variance risk across maturities and different market regimes over the period of January 2000 to July 2025. Following the second-order polynomial Implied volatility curve parameterisation of Zhang and Xiang (2008), the implied volatility smirk is decomposed into its level, slope, and curvature factors at six constant maturities spanning 30 days to 360 days.

We extend the work of Guo et al. (2003) and implement a regime-conditional regression framework to investigate whether the term spreads can predict the variance risk premium across contango and backwardation regimes. Additionally, we introduce a rate-normalised hump factor to capture the non-linear medium term curvature dynamics beyond end point based spreads.

Using At-the-money straddle returns and synthetic variance swap returns as model-free proxies of the variance risk premium; we find that the short-end level spread between 30 to 180 days significantly predicts the monthly straddle returns at 3 months through to 6-month maturities. The regime-conditional indicator reveals a complete sign reversal in long-end spread's predictive slope at 12 months maturity the two regimes in the in-sample regression. The level hump factor provides supplements in-sample evidence at medium to long maturities. Out of sample, the long-end spread between 360 day and 180-day implied volatility is the most robust predictor at the 12-month horizon, while the hump factor does not generalise beyond the estimation period. These findings suggest that structural assumptions of long run variance provide the most reliable and persistent signal for predicting the variance risk premium.

1. Introduction

The Chicago Board Options Exchange (CBOE) introduced the original volatility index (VXO) in 1993 to track Standard & Poor's (S&P) 100 at-the-money implied volatility based on the Black-Scholes model (Bonaparte et al., (2023)). In 2003, the CBOE redefined the index in collaboration with Goldman Sachs, switching the underlying tracking benchmark to the S&P 500 index (VIX). This simultaneously adopted a model-free approach derived from the prices of a portfolio of 30 calendar day options with weights inversely proportional to the squared strike price (Fernandes et al., 2014). This approach replicates a log-contract to synthesise the risk-neutral expected variance of the largest publicly traded companies in the United States under the equivalent martingale measure. Jiang and Tian (2005) empirically show that this replicating portfolio requires no assumption about the underlying return generating process, which is precisely what allows VIX to be interpreted as a model-free measure of expected variance rather than a parameter fitted to a specific pricing model. Since then, it has become the benchmark for United States (US) equity markets as it reflects the conditional expectation of future S&P volatility under the physical measure. Beyond its utility as a headline fear gauge of market sentiment, the VIX underpins a vast, highly liquid ecosystem of tradable volatility products which including direct derivatives, such as VIX futures and options.

The gap between what VIX implies and what subsequently materialises is not merely a statistical curiosity, but a tradable commodity. As demonstrated by Bordonado et al. (2016), this gap is actively monetised by market participants to drive price discovery, evaluate portfolio hedging constraints, and execute tactical volatility trading strategies. Utilizing VIX derivatives, Cheng (2019) demonstrates that VIX futures carry significant, time-varying premium that accounts for future conditional volatility. Bakshi and Madan (2006) predict positive volatility spreads when investors are risk-averse, and when the physical index distribution is negatively skewed. Similarly, Bakshi and Kapadia (2003) provide empirical evidence that supports the negative market volatility risk premium theory. Consequently, this implies the VIX systematically overstates realised volatility and suggests assets with positive variance risk exposure earn negative abnormal returns. Most studies focus on the return predictability of Variance Risk Premium (VRP), such as Bollerslev et al. (2009), which demonstrate that the ex-ante VRP predicts aggregate equity returns with substantial forecasting power over the immediate quarterly horizons, measured as the spread between the model-free implied and realised variance. Due to the ex-ante VRP being unobservable directly, empirical work must proxy it either by evaluating the spread between option-implied variance and subsequent realised variance, or by tracking the ex-post returns of variance-sensitive portfolios, such as variance swaps and delta-neutral straddles, whose returns are approximately proportional to volatility innovations (Cremers et al. 2016).

A comparatively minor strand of literature investigates the inverse of the question, which predicts the variance risk premium itself. Konstantinidi and Skiadopoulos (2016) investigate this by distinguishing the investment horizon from the variance swap's maturity, evaluating VRP as the conditional expectation of returns on a position that can be closed prior to expiry. Results showed that trading activities, and the resulting VRP-conditioned trading strategy, outperform benchmarks after transaction costs. However, these proprietary over-the-counter variance swap quotes limit transparency and replicability.

Guo et al. (2023) addresses this by examining whether the term structure of the S&P 500 implied volatility (IV) smirk predicts the variance risk premium. The IV smirk is deconstructed into a second-order polynomial to describe the level, slope and curvature, following a local expansion method proposed by Zhang and Xiang (2008). The variance-sensitive asset returns used in the paper were derived from data developed through synthetic S&P 500 variance swaps, futures contracts and straddle option positions by Johnson (2017). According to the paper, this approach avoids measurement issues inherent in estimating realised variance from historical index returns, which are typically required to construct the physical side of the standard implied-minus-realised VRP measure. The paper found that the term spread of the implied volatility level factor significantly predicts the VRP in both in-sample and out-of-sample tests, with greater predictability at a one-week horizon. However, this assumes the directional

movement of the implied volatility term structure is constant. Christoffersen et al. (2009) show that the level and slope of the index option smirk fluctuate independently of each other, and that single-factor models are consequently unable to capture the full richness of the smirk's evolution across maturities. If the middle of the curve moves independently of its two ends, a hump or trough could be forming that a simple endpoint-based spread does not account for.

Following the framework developed by Zhang and Xiang (2008), we aim to define the shape of the implied volatility curve in terms of its level, slope, and curvature. These factors are approximately equivalent to the VIX standard deviation, skewness and excess kurtosis under the risk-neutral moments formula developed by Bakshi, Kapadia and Madan (2003). In this paper, we aim to investigate whether the implied volatility curve rising in a front-loaded or a back-loaded curve carries additional predictive information for the VRP. This is achieved by decomposing the total term spread into short-end (180-to-30 days) and long-end (360-to-180 days) segments which allows us to map these non-linear dynamics to distinct transient spikes and persistent volatility risk expectations (Mixon, 2007; Todorov, 2010; Drechsler, 2013). The VRP data for this study implements variance-sensitive assets namely delta-neutral straddle portfolio returns following the methodology implemented in Coval and Shumway (2001), and synthetic variance swap contracts as in Dew-Becker et al. (2017).

Additionally, this study investigates whether the predictive power of the term spread is conditional on the regime in which it is observed, rather than stable across regimes. We define the regimes as either contango, characterised by an upward-sloping IV curve where investors demand variance-hedging instruments in anticipation of rising future volatility, and backwardation, where crisis periods spike near-term IV above long-term levels before volatility mean reverts toward long-run steady state. We address this by introducing a regime-conditional indicator dummy variable, $D_t \in \{0,1\}$, where $D_t = 0$ denotes positive term spread and $D_t = 1$ denotes negative term spreads. This allows both the intercept and the slope of the predictive regression to vary across regimes. This provides a direct test on whether the term spread retains statistically significant predictive power for the VRP in both states, rather than assuming the single-regime relationship as in Guo et al. (2023). By better understanding the predictive power of information incorporated within the VIX term structure, allowing investors to manage variance risk during crisis periods or calm periods.

While methodologically sound, Guo et al.'s (2023) analysis is confined to a sample period that predates several of the most impactful volatility regimes of the past decade. This is because the variance-sensitive asset return data from Johnson (2017) are limited to 1996 – 2017. Events such as the “Volmageddon” episode which refers to the abrupt market crash of short volatility strategies on 5th February 2018 (Augustin et al., 2021), the COVID-19 volatility shock (Apergis

et al., 2023), the 2022 monetary tightening cycle and the rise of zero-days-to-expiration options (Brogaard et al., 2023). According to research (Tong et al., 2022), they found that volatility risk aversion was time-varying and identified specific low-volatility periods: 2004-2007 and 2014-2017 versus other periods dominated by the high VRP regime. This paper extends the sample periods from 2000 to 2025 to account for these volatility spikes.

We find that the short-end level spread significantly predicts monthly straddle returns at medium maturities in sample. The regime-conditional framework reveals a complete sign reversal in the long-end spread's predictive slope at the 12-month maturity between backwardation and contango regimes. This indicates that a single-regime specification cannot fully investigate the relationship between term structure and the VRP. Out of sample, the long-end level spread achieves a Clark-West statistic of 2.707 ($p=0.003$) at the 12-month horizon. This establishes long-run variance dynamics of VIX as a structural feature of the index. The hump factor provides supplementary in-sample evidence at medium maturities but fails to generalise out-of-sample, indicating period specific volatility dynamics.

The remainder of this paper is organised as follows: section 2 provides a literature review on the economics and asset pricing of the variance risk premium to establish the foundations of the risk premia as a reward for bearing volatility risk

and outlines methods used to capture these unobservable expectations; and the geometry and factorisation of the implied volatility smirk to explain how it is synthesized and decomposed. Section 3 elaborates the data and methodology used in the study including a description of the regression analysis employed in the study. Section 4 provides an analysis of the empirical results of the study including the necessary robustness tests to validate findings and finally section 5 provides the conclusion and discussion on the limitations of this study.

2. Literature review

2.1. The economics and asset pricing of the variance risk premium.

The variance risk premium (VRP) is not the direct compensation for holding equity risk itself, but rather the reward investors require to justify bearing exposure to the unpredictable fluctuations in market volatility over time. It is formally defined as the gap between the market's risk-neutral expectation of future variance, and the realised variance. Beyond independent volatility shocks, return variance can vary stochastically based on multiple theoretically distinct reasons. This can be through its correlation, with the underlying assets stock price or return, as captured by constant elasticity of variance model investigated by Beckers (1980) who find an inverse relationship between a company's stock price and its return volatility. The variation in returns can also be attributed to independent fluctuations as a distinct source of risk unrelated to price movements, as in the stochastic volatility models of Heston (1993).

For the S&P 500, a closely related measure is the VIX premium, defined as the difference between the risk-neutral and physical expectations of the future VIX index level as in Cheng (2019). This premium is non-zero echoing the foundational equilibrium framework from French et al. (1987), who find that the expected market risk premium is positively related to the predictable component

of stock market volatility, driven by volatility clustering, providing early evidence that investors demand greater returns in anticipation of rising volatility. Recent work by Feunou et al. (2017) provides further evidence of a positive and significant link between the downside VRP and the equity premium by deconstructing volatility based on the behavioural assumption that investors view it asymmetrically. According to research, this negative unconditional volatility and risk premia is widely attributed to risk averse market participants' demand for volatility insurance to hedge equity portfolios against severe market drawdowns (Carr and Wu, 2009; Bakshi and Kapadia, 2003).

Under Epstein-Zin-Weil recursive preferences calibrated with standard preference for the early resolution of time-varying uncertainty, an agent's marginal utility, is determined by their lifetime utility (Epstein and Zin, 1991). As heightened future volatility reduces lifetime utility, investors exhibit a strong intertemporal hedging demand to protect against shocks to future expected volatility. The VRP emerges as a priced factor whereby its temporal fluctuations can be captured by incorporating a time-varying volatility-of-volatility state variable (Bollerslev et al., 2009; Barndorff-Nielsen and Veraart., 2012). Additionally, the VRP represents compensation for highly persistent, expected consumption growth prospects and fluctuating economic uncertainty according to Drechsler and Yaron (2011), who extend the long-run risk model of Bansal and

Yaron (2004) to connect the option-implied risk premium to a structural model of macroeconomic fundamentals and investor preferences.

In predictive finance and options literature, the conventional approach for deriving the VRP introduces a profound set of statistical, econometric and observational challenges. This is due to a timing mismatch between spot market volatility shocks, which are observed in real time through implied volatility, and their eventual realisation as observed variance resulting in difficulty when simultaneously measuring them at a given time to extract the risk premia (Todorov, 2010). Some studies overcome this by relying on ex-post (after the event) realised variance as a substitute for the ex-ante (before the event) expected variance based on the assumption that on average, realised variance is an unbiased proxy of investor expectations. This results in measurement and estimation noise due to realised variance containing high-frequency price jumps and sampling errors that do not reflect ex-ante expectations as highlighted in Bollerslev et al. (2009). Alternatively, the conditional expectation of variance can be estimated using historical parametric forecasting models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) or ARMA (Autoregressive Moving-Average) as in Liu et al. (2015). However, they fail to capture differing rates of mean reversion and are disconnected from economic fundamentals as they rely on a single persistence parameter. Modern studies address the term-structure dynamics by applying advanced multi-factor pricing models, such as the three-factor self-exciting stochastic volatility model proposed by Li and Zinna

(2018). However, such frameworks are highly computationally intensive requiring parallel computing to filter and estimate latent state variables.

To bypass these hurdles, instead of estimating the unobservable expectations of variance under the physical measure, several studies have focused on tracking the ex-post realized returns of variance-sensitive assets such as delta-neutral straddles or synthetic variance swaps. Due to delta-hedging removing the portfolio's directional exposure, these assets function as direct, model-free empirical proxies for the realized variance risk premium whereby their instantaneous returns reflecting volatility innovations under stochastic volatility allowing for their use in tradeable strategies (Guo et al., 2023; Cremers et al., 2016; Johnson., 2017). Additionally, to investigate the information content embedded in the S&P 500 implied volatility (IV) term structure regarding the risk premia at varying horizons (30, 60, 90, 180 and 360 days), the empirical design must align the daily option-implied predictors with corresponding constant maturity VRP proxies. Implementing this daily, multi-maturity framework via constant-maturity interpolation is imperative to prevent maturity contraction biases and avoids discrete roll-yield distortions enabling continuous, daily time series measures suitable for predictive regressions according to Konstantinidi and Skiadopoulos (2016).

2.2. The geometry and factorisation of the option smirk

The extraction of the risk-neutral, expected future variance directly from option prices is based on the framework established by Britten-Jones and Neuberger (2000). They show that for a given underlying asset, all continuous price processes consistent with option prices share a common expectation of squared return volatility over a given horizon. This framework demonstrates that a portfolio of Out-of-the-money European puts and calls, paired with dynamic hedging strategies in the underlying asset, can perfectly replicate risk-neutral expected integrated variance. This builds on the foundational work of Breeden and Litzenberger (1978) who show that prices of primitive securities can be extracted from the market prices of European call options without requiring restrictive assumptions about price diffusion processes. They show that the second derivation of call price, with respect to strike price, recovers the risk-neutral probability density of the underlying asset at expiration. The result is a model-free implied volatility formula that extracts the forward-looking expected variance directly from observable market prices without requiring parametric model specifications or assumptions on constant volatility. Jiang and Tian (2005) extend this framework to general jump-diffusion settings by leveraging martingale decomposition, proving that the spanning relationship between option prices and expected integrated variance remains robust even in the presence of price discontinuities.

Building on this insight, Bakshi, Kapadia and Madan (2003) derive model-free formulas for the risk-neutral variance, skewness, and excess kurtosis of returns. They define the quadratic, cubic and quartic contracts respectively on log returns given a portfolio of out-of-the-money calls and puts weighted inversely by their squared strike price without imposing a parametric structure on the pricing kernel or specifying the stochastic process of the underlying volatility. This allows for clean distinguishing between the pricing of continuous volatility risk and the pricing of discontinuous, asymmetric crash risks. Sven Balder (2015) mathematically reconciles the differences between the expected integrated variance methodology with the actual risk-neutral variance of log-returns spanned by the moment's contracts of Bakshi and Kapadia (2003). He proves that the actual variance of log-returns, reflected by the first and second moments of risk-neutral return distribution, incorporates three additional risk factors omitted by expected quadratic variation developed by Britten-Jones and Neuberger (2000) namely, the variance of the integrated variance process, the covariance between the integrated variance and the continuous diffusion component of the underlying stock prices as well as the jump components of stock prices.

The Bakshi-Kapadia-Madan (BKM) approach requires a sufficiently dense cross-section of strike prices at each maturity to approximate the underlying integral describing the implied volatility surface which limits its applications to only

highly liquid option markets. Zhang and Xiang (2008) propose a complementary characterisation as an alternative industry standard to quantify the risk-neutral distribution by directly parameterizing the shape of the implied volatility smirk. At a fixed maturity, they achieve this by developing a second-order polynomial fitted to the implied volatility function over a standardized moneyness measure. This moneyness measure, defined as the logarithm of the strike price for the options over the implied forward price, is determined based on put-call parity. The authors normalise the equation by the standard deviation of returns to that maturity as in Carr and Wu (2003), who implement a similar constant average volatility as a denominator. The factorisation decomposes the smirk into three constant maturity factors for each day resulting in the level, slope and curvature coefficients which are approximately related to the risk-neutral moments. This is achieved by matching the two option-pricing formulas At-the-money (ATM), their first order, and second-order sensitives with respect to their respective strike prices. Specifically, by matching option prices via the Black-Scholes formula using the developed implied volatility function and the risk-neutral expectation formula evaluated under the Edgeworth expansion of the probability density with known skewness (SKEW) and excess kurtosis (KURT). By enforcing that the two frameworks provide the same values for the at-the-money option, the authors show that ATM volatility level represent the volatility level of the underlying asset; the first order derivation related to a risk reversal portfolio establishing the

slope; and the second-order sensitivity relates to a butterfly spread portfolio establishing the curvature.

Given traditional parametric models require assuming a continuous-time process for the underlying asset price and volatility, such as the single-factor stochastic volatility model under two-factor diffusion of Heston (1993) or the multi-factor stochastic volatility model of Christoffersen et al. (2009), by directly parameterising the observable implied volatility curve shape, the framework developed by Zhang and Xiang (2008) capture changing market expectations without assuming a structural return-generating process. This allows for model-free calibrations without severe estimation errors, such as non-linear optimization instabilities or parameter drift over time as highlighted by the ten-parameter time changed Levy process of Huang and Wu (2004) and Bates (2022) respectively. Additionally, as the Zhang and Xiang framework requires only three out-of-the-money option prices per maturity to fit the implied volatility surface, this allows researchers to estimate the implied level, slope and curvature factors using sparse options data. Gehricke and Zhang (2019) implement the exact framework of Zhang and Xiang (2008) to document and analyse the implied volatility curves of the iPath Series B S&P 500 VIX short-term futures ETN (VXX). Given VXX options data is limited compared to that of VIX as highlighted by trading volume and open interest as highlighted in (Basta and Molnar, 2018), the flexibility of Zhang and Xiang framework to be implemented effectively regardless of this constraint is highly beneficial.

3. Data and methodology

3.1. Implied Volatility curve factors

To calculate the level, slope and curvature factors, we obtain S&P 500 (SPX) options data from Ivy DB OptionMetrics via Wharton Research Data Services (WRDS), from January 1, 2000 to July 31, 2025. Optionmetrics provides option price data on daily implied volatility, open interest, type of option contract, best bid, best offer, date issued, expiry date, volume, strike price, and AM-settlement type. We follow the options data cleaning methodologies, as in Neumann and Skiadopoulos (2013):

- We consider only out-of-the-money (OTM) and at-the-money (ATM) options.
- We discard options with zero bid price and those that have a mid-price of the best bid and offer less than 0.2 \$.
- We exclude options with a maturity of fewer than 7 days to exclude zero-days-to expiry options.
- We discard options without implied volatility values.
- We discard options with zero open interest and or zero trading volume.
- We remove contracts with less than three strike prices on a given day to calculate the implied volatility curve.

Table 1 presents the summary of filtered and cleaned SPX options trading data grouped by maturity category ranging from 30, 60, 90, 120, 180, and 360 days. From the table, we can see that open interest, representing the number of contracts outstanding, is highest in the 30 to 90-day bucket corresponding to 31.93 % of the overall sample data. Trading volume declines with increase in maturity, falling from 223,309.06 contracts at maturities below 30 days to 11,056.90 at maturities greater than 360 days. This likely reflects participants using SPX options for near-term hedging, speculation, or tactical position. The coefficient of variation (CV) for trading volume rises from 68.61 % at the shortest tenor bracket to 99.53% at maturities greater than 1 year, indicating a 45.1% increase in relative dispersion. This indicates that long-dated trading activity is characterised by limited volume on most days with infrequent large transactions rather than consistent liquidity. Open interest CV remains relatively stable across maturities, suggesting positions persist independent of day-to-day transactions. Strike availability peaks at 179.21 strikes per day in the 30–90-day bucket before declining to 78.30 at maturities beyond 360 days likely capturing the substantial expansion in listed strikes over the sample period. This liquidity profile implies that risk premium returns estimated at the 12-month maturity (>360 days) carry substantially greater measurement uncertainty than at short term maturities thus enforcing the need to investigate the hump dynamics at the medium-term maturities (180 days) as stated in section 3.3.

Table 1 Summary of trading activity grouped by maturity for S&P 500 index options.

	Overall	<30	30-90	90-180	180-360	>360
mean open interest	6996251.05	1762070.69	2234126.18	1277501.92	1145346.98	736708.72
std dev open interest	3459120.97	1070271.01	1405997.80	842855.95	743721.82	446572.59
CV open interest	49.44	60.74	62.93	65.98	64.93	60.62
mean open trading volume	479710.68	223309.06	183247.44	53003.95	29107.26	11056.90
std dev trading volume	297912.06	153203.77	135010.91	48496.80	26674.39	11005.29
CV trading volume	62.10	68.61	73.68	91.50	91.64	99.53
mean strike number	216.56	147.01	179.21	138.32	82.85	78.30
std dev strike number	148.05	118.15	137.08	132.14	60.26	50.99
CV strike number	68.37	80.37	76.49	95.53	72.73	65.12

Table 1 reports summary statistics for cleaned and filtered Standard & Poor's Index (SPX) option data sourced from WRDS grouped by five maturity buckets spanning maturities below 30 days, between 30 to 90 days, between 90 to 180 days, between 180 to 360 days, and exceeding 360 days covering the full sample period from 2000 to 2025. Three dimensions of market activity are reported namely open interest, trading volume and number of available strike prices each summarised by daily mean, standard deviation and coefficient of variation (CV).

3.2. Quantifying the implied volatility curves

We follow the local expansion method proposed by Zhang and Xiang (2008) to quantify the shape of the implied volatility curve for each maturity across the moneyness function. The curve is first parameterized locally by a quadratic function of the standardized moneyness variable. It is then mapped via Breeden and Litzenberger (1978) identity into a risk-neutral distribution. The coefficients of the quadratic equation, in closed form, are then related to the standard deviation, skewness and excess kurtosis of that risk-neutral distribution.

Consider a portfolio constructed by selling one put and buying one call option, each with the same strike price and exercise date where the payoff equals the forward price. Let $c_0(K)$ and $p_0(K)$ denote the market prices of the call and put option with the same maturity, τ . The implied forward price F_0 is obtained through the put-call parity relationship, evaluated at the strike price that minimizes the absolute call-put price difference:

$$F_0 = K^* + e^{r\tau} (c_0(K^*) - p_0(K^*)) \quad (1)$$

$$\text{where } K^* = \arg \min_K |c_0(K) - p_0(K)| \quad (2)$$

Rather than working directly with strike price, Zhang and Xiang (2008) suggest using the moneyness variable, ξ , defined as the logarithm of the strike price over the implied forward price scaled by the standard deviation of the return on maturity following the industry notation by Carr and Wu (2003) when using a constant benchmark volatility,

$$\xi = \frac{\ln\left(\frac{K}{F_0}\right)}{\sigma\sqrt{\tau}} \quad (3)$$

Where σ is the standard deviation, proxied by the VIX index, F_0 is the forward price, K is the strike price and τ is the time to maturity.

At a fixed maturity, a second-order polynomial is used to describe the implied volatility as a function of the moneyness function:

$$IV(\xi) = \gamma_0(1 + \gamma_1\xi + \gamma_2\xi^2) \quad (4)$$

Where γ_0 , γ_1 , and γ_2 are the level, slope, and curvature factors of the implied volatility smirk. This method allows the use of three IV points to accurately fit the curve on each day for a particular maturity.

Specifically, the at-the-money volatility level establishes the baseline spot volatility of the underlying asset (γ_0). Regarding the slope coefficient, this mirrors the payoff of a risk-reversal portfolio which comprises of a long position on out-of-the-money call options and a short position of out-of-the-money put options on the underlying asset. Finally, the curvature coefficient mirrors the payoff of a butterfly spread portfolio which comprises of three positions namely a long position on two call options, whereby one is in-the-money and the other is out-of-the-money, and a short position on two at-the-money call options.

3.3. Constant maturity factors and term spreads

Given that on each day's Implied Volatility (IV) smirk is fit independently for the given days to maturity quoted on that day, the resulting level, slope and curvature factors are not directly comparable across dates. Therefore, we examine how the shape of the smirk evolves along the term structure over time through linear interpolation of the IV factors to the constant maturities we are investigating. Matching the CBOE VIX-style term structure conventions, we obtain constant maturity factors at 30, 60, 90, 120, 180 and 360 days represent 1 month, 2 months, 3 months, 4 months, 6 months and 1 year to maturity as demonstrated by Lin (2013).

Extrapolating the fit beyond the observed range risks producing unstable and unrepresentative values. To safeguard against this, when the target maturity is beyond the available maturity range, the IV factors are set to be the upper and lower bound of the factors within the maturity group. On a given day, we form the factor term spreads of the IV curves by deriving the difference between the high and low IV factors set as 360 days and 30 days respectively for the level, slope and curvature as follows:

$$\gamma_0^{360-30} = \gamma_0^{360} - \gamma_0^{30} \quad (5)$$

$$\gamma_1^{360-30} = \gamma_1^{360} - \gamma_1^{30} \quad (6)$$

$$\gamma_2^{360-30} = \gamma_2^{360} - \gamma_2^{30} \quad (7)$$

Where γ_0 , γ_1 , and γ_2 are the level, slope, and curvature factors of the implied volatility smirk. The term spread is set as between 360 days to 30 days for each of the factors. In addition, we consider 180-days to 30-day spread to verify the predictability of returns as a shorter spread representing mid - high to low days to maturity.

According to Christoffersen et al. (2009), the volatility term structure is not necessarily monotonic in that its directional movement remains unchanged therefore It suggests a single factor model does not capture the significant curvatures across maturities. Todorov (2010) investigated the dynamics of variance risk premium and found that investors' fear of future jumps (short-term volatility spikes) is sensitive to recent jump activity resulting in increased purchase of variance-sensitive assets to hedge against volatility following immediate occurrences of jumps.

Therefore, we include a middle vs middle spread diagnostic of genuine curve dynamics rather than just endpoint level using a discrete second deference on three nested points being short, middle and long as follows:

$$H_0 = \frac{[\gamma_0^{360} - \gamma_0^{180}]}{180} - \frac{[\gamma_0^{180} - \gamma_0^{30}]}{150} \quad (8)$$

$$H_1 = \frac{[\gamma_1^{360} - \gamma_1^{180}]}{180} - \frac{[\gamma_1^{180} - \gamma_1^{30}]}{150} \quad (9)$$

$$H_2 = \frac{[\gamma_2^{360} - \gamma_2^{180}]}{180} - \frac{[\gamma_2^{180} - \gamma_2^{30}]}{150} \quad (10)$$

Where H_0 , H_1 , and H_2 represent the level, slope and curvature hump. We rate normalised each half by its span before differencing convert into a per day rate

change. This accounts for the overall difference between each half in terms of days to ensure we investigate equal sections.

1. If the curve is relatively linear with no hump, the two halves are similar or cancel each other out resulting in $Hump_{0,1,2} \cong 0$.
2. If the curve bulges upward in the middle and flattens toward the long end i.e. $[\gamma_{0,1,2}^{180} - \gamma_{0,1,2}^{30}] > [\gamma_{0,1,2}^{360} - \gamma_{0,1,2}^{180}]$, this results in $Hump_{0,1,2} < 0$.
3. Conversely, if the curve bulges downward in the middle and flattens toward the long end i.e. $[\gamma_{0,1,2}^{180} - \gamma_{0,1,2}^{30}] < [\gamma_{0,1,2}^{360} - \gamma_{0,1,2}^{180}]$, this results in $Hump_{0,1,2} > 0$.

To account for outlier predictors on days with thin quoting or unstable fits, we winsorize the level, slope, and curvature term spread factors as well as the hump at the 0.5% tail cutoff of the whole sample adopting the framework of Cao and Han (2013).

3.4. The Variance Risk Premium data

In this study, we apply synthetic variance swap contracts and S&P 500 straddle option portfolios returns as proxies for the variance risk premium. This approach avoids the measurement error inherent in computing realised variance directly from historical index returns. The VRP data extracted is relevant to the level factor as it corresponds to the risk-neutral variance of the VIX implied volatility surface.

3.4.1. S&P 500 straddle returns

An option straddle portfolio is defined as a strategy that combines a long position in both call and put options respectively where both options share the same maturity and strike price. Empirical research has shown that long positions in at-the-money straddles consistently yield negative returns on average regardless of actual volatility movements consistent with the variance risk premium (Arisoy, 2010; Barinov, 2008).

We follow the methodology as in Coval and Shumway (2001) to derive the returns of straddle positions as detailed in the Appendix. Options on the S&P 500 are grouped by their moneyness level, with at-the-money (ATM) options defined as the difference between the underlying asset price and the strike price within the range of zero. The individual option returns are then calculated with the overall straddle return as the equal weighted average of returns from ATM call

and put options. Daily straddle returns are then cumulated into monthly returns to serve as the basis for empirical tests.

3.4.2. synthetic variance swaps

A variance swap is an over the counter (OTC) financial derivative where two parties agree to exchange cash flows at maturity based on the volatility of a set underlying asset. Within this contract, one party pays a fixed rate defined as the variance swap price set at the start of the contract. In return, they receive a floating payment equal to the actual realized variance of the given underlying asset accumulated over the contract's life.

To replicate this derivative over the set sample period, we develop synthetic variance swap contracts designed to swap a fixed payment for a variable amount proportional to the realised variance of the S&P 500 index based on the work of Carr and Wu (2008). We follow the methodology employed by Dew-Becker et al. (2017) using a deep cross-section of S&P 500 index options. The variance swap payoff replicated by holding a static, continuously weighted portfolio of out-of-the-money (OTM) options comprising of both put and call options across a continuum of strike prices expiring on the swap's maturity date as detailed in the Appendix.

3.5. Regime-conditional regression analysis

The baseline regression analysis developed by Guo et al. (2023) investigated whether the shape of the implied volatility term structure could predict the variance asset returns by imposing a single, time-invariant coefficient β as below:

$$R_{t:t+1} = \alpha + \beta X_t + \epsilon_{t:t+1} \quad (11)$$

Where $t = 1, 2, \dots, T - 1$, $R_{t:t+1}$ is the next-day, next-week, or next-month asset returns following the subscript convention from Rapach et al. (2016). X_t is the level, slope or curvature of term spread. This assumes a constant relationship between each predictor and subsequent returns across the sample period whereby a positive term spread signals that investors expect future volatility to rise, prompting increased demand for variance-hedging instruments. This in turn bids up the price of said instruments leading to negative returns for buyers of variance-sensitive assets represented by the negative variance risk premium. The direct economic interpretation of this relationship is referred to as contango, which is a state when the market prices in calm conditions in the present but expects more uncertainty in the future leading to the upward sloping IV curve across maturities. Mixon (2007) found that mean reversion was the key feature of the IV term structure with short-dated IV reacting sensitively to current conditions, while long-dated IV stayed comparatively stable, reflecting expectations of volatility

converging towards long-run levels. Specifically, buyers of variance protection overpay for these assets relative to realised volatility thus the VRP is negative from the buyer's perspective and positive from the seller's perspective.

Conversely, a negative term spread where short-dated IV exceeds long-dated IV, produces a downward-sloping curve interpreted as backwardation, which is a state when the market is already stressed but expects conditions to normalize over the long term consistent with the mean-reverting dynamics of variance modelled by Zhang, Shu and Brenner (2010). The transmission mechanism operating in this regime runs in the opposite direction to contango. As volatility shocks are already materialised, hedging demand is front loaded rather than anticipated leading investors to seek immediate protection rather than pricing in future rises. To investigate these distinct states, we introduce a regime-conditional indicator:

$$D_t = \begin{cases} 1, & \text{negative term spread (backwardation)} \\ 0, & \text{positive term spread (contango)} \end{cases} \quad (12)$$

By formalising the backwardation and contango state, we conduct in-sample and out of sample tests on next-day, next-week and next month horizons to test whether the relationship between term spreads and variance risk premium varies systematically with the prevailing market regime with a statistically significant coefficient indicating regime-dependence.

3.5.1. In-sample analysis

We formulate the following regression to examine whether the implied volatility factors can predict the variance-sensitive asset returns across different maturities with consideration to the distinct regimes given by:

$$R_{t:t+1} = \alpha + \beta_1 X_t + \beta_2 D_t + \beta_3 X_t D_t + \epsilon_{t:t+1} \quad (13)$$

Where D_t is the regime indicator, $t = 1, 2, \dots, T - 1$, $R_{t:t+1}$ is the next-day, next-week, or next-month asset returns, and X_t is the level, slope, or curvature term spreads or hump factors for each regression.

We run overlapping regressions at each horizon whereby the next-week returns compound the next-day returns from $t + 1$ to $t + 5$ representing a trading week from Monday to Friday, and the next-month return compounds next-day returns from $t + 1$ to $t + 21$ representing four trading weeks, consistent with standard practice in predictive return literature (Jondeau et al., 2019). Standard errors (SE) from the regression analysis are adjusted to make them more reliable and robust using Newey-West heteroscedasticity and autocorrelation consistent estimators (Newey and West, 1987) with 4 lags for weekly and 20 lags for monthly returns, given the persistence of the daily IV curve factors. This adjusts the SE of the regression coefficients such that hypothesis testing remains asymptotically valid

regardless of whether non-constant variance and serial correlation are present in the residuals. If the coefficients of the factors are statistically significant with a $p\text{-value} < 0.05$ with the Newey-West adjusted t-statistic compared against the standard normal distribution for a two-tailed test, this indicates that the term spreads or hump factor carries genuine predictive information for the future variance risk premium rather than reflecting noise in fitted IV curve. Table 2 below shows the interpretation of results from the regression analysis.

TABLE 2 Interpretation of linear regression coefficients.

Regime	intercept	slope
Contango	α	β_1
backwardation	$\alpha + \beta_2$	$\beta_1 + \beta_3$

Table 2 reports the regime-specific parameters within the in-sample regression equation. As the indicator variable takes the value zero in contango, that regime's intercept and slope are given directly by α and β_1 respectively. In backwardation, the indicator takes the value of one thus the effective intercept becomes $\alpha + \beta_2$ and the effective slope becomes $\beta_1 + \beta_3$.

Given during backwardation, β_2 represents the intercept shift which indicates whether the average level of returns is different across regimes independent of the value of the spread itself and β_3 represents the slope shift which indicates whether a one-unit change in spread moves returns by a varied amount depending on regime. A statistically significant β_3 indicates the predictive relationship is not a single global effect that is monotonic, but instead differs meaningfully in strength and potentially in sign across regimes. Specifically, if β_3 is negative and

β_1 is negative, the total slope in backwardation is more negative suggesting term spreads predict more strongly the negative Variance risk premium returns during stress. Conversely, if β_3 is positive and large enough to flip the sign of $\beta_1 + \beta_3$, this suggests the relationship reverses in backwardation.

3.5.2. Out-of-sample analysis

In section 3.5.1. we conduct the regression analysis whereby the full 2000- 2025 sample is partitioned into a 15 year in-sample estimation period dated January 2000 – December 2015 and an approximately 10 year out-of-sample evaluation period dated January 2016 – July 2025. This follows the conventional 60/40 temporal split applied in return predictability literature (Campbell and Thompson, 2008). The out-of-sample evaluation is conducted on data withheld entirely from the estimation stage, with the split point fixed prior to model estimation to preclude the look-ahead bias. The out-of-sample period encompasses at least three structurally distinct volatility regimes namely the low-volatility expansion of 2016-2019, and the extreme volatility spike caused by the XIV and SVXY products collapse known as the “Volmageddon” episode as investigated by Augustin et al., 2021. The COVID-19 volatility shock of 2020, and the 2022 monetary tightening cycle during rate-driven bear markets in the US equities are also included. These additional volatility regimes provide a reliability test of the model’s generalisability.

Similar to the in-sample tests, the out-of-sample analysis is run over the same next-day, next-week, and next month variance asset returns. The in-sample estimated coefficients are then applied to the out-of-sample observations of X_t and D_t to generate one-period-ahead forecasts like those of the synthetic variance swap and straddle portfolios, as in the expression below:

$$\hat{R}_{t:t+1} = \hat{\alpha} + \hat{\beta}_1 X_t + \hat{\beta}_2 \overline{D}_t + \hat{\beta}_3 X_t \overline{D}_t \quad (14)$$

Where $\hat{\alpha}$, $\hat{\beta}_1$, $\hat{\beta}_2$, and $\hat{\beta}_3$, are the coefficients based on the regression from 2000-2015 data and held fixed for out-of-sample forecasts, $\overline{D}_t$ is the regime indicator, $\hat{R}_{t:t+1}$ is the predicted return on date $t + 1$ using the coefficients from the in-sample regression. To compare the model's forecast error against the prevailing mean forecast as the benchmark being the historical mean computed from the in-sample period, we employ the out-of-sample R^2 as the primary metric for evaluation in line with Campbell and Thompson (2008) given by:

$$R_{OS}^2 = 1 - \frac{\sum_{t=t_0}^{t_n} (R_{t+1} - \hat{R}_{t+1|t})^2}{\sum_{t=t_0}^{t_n} (R_{t+1} - \bar{R}_{IS})^2} \quad (15)$$

Where R_t is the real return at time t , $\hat{R}_t$ is the predicted return at time t which is the model's one-period ahead forecast, and $\bar{R}_{IS}$ is the in-sample historical mean

from the beginning through time t . A positive R_{OS}^2 indicates the model outperforms the historical mean with the inverse being true.

As we are comparing a restricted model based on the historical mean, against a nested unrestricted model defined by the regression analysis, we apply the Clark-West Mean squared predicted error-adjusted statistic (MSPE) to correct for this (Clark and West, 2007). This measures how far the regression analysis' predictions are from the actual observed values as defined as:

$$\mathcal{F}_{t+1} = (R_{t+1} - \bar{R}_{IS})^2 - \left[(R_{t+1} - \hat{R}_{t+1|t})^2 - (\bar{R}_{IS} - \hat{R}_{t+1|t})^2 \right] \quad (16)$$

Where $\bar{R}_{IS}$ is the forecasting value for the prevailing mean forecast model, and $\hat{R}_{t+1|t}$ is the forecasting value from the specified factor model. We compute $\bar{\mathcal{F}} = T^{-1} \sum_t \mathcal{F}_{t+1}$ which is the MSPE that is tested on whether it is significantly positive using the t-statistic from regressing $\mathcal{F}_{t+1}$ on a constant with Newey-West standard errors (Newey and West, 1987). As an additional evaluation metric, we report directional accuracy (DA), defined as the relative consistency in which the predicted sign of the variance-sensitive asset returns matches between in-sample and out-of-sample regression analysis. Significance is assessed using a one-sided binomial test against a null requiring a 50% hit rate, evaluated between the weekly and monthly observations.

4. Empirical results

Table T1 compares the Implied forward price obtained from put-call parity at the strike minimising the absolute call-put difference against an Ordinary Least Squares put-call parity benchmark across 148,194 observations from January 2000 to July 2025. They differ by a mean of 0.88 basis points across all maturities with divergence linearly rising with the maturity horizon from 0.49 basis points at maturities less than 30 days to 2.29 basis points above 360 days. This is consistent thinner strike coverage at longer maturities as in Table 1, whereby fewer near-the-money quotes anchor either estimator. Therefore, the choice of forward price estimator has no material effect on the Implied Volatility factors or Variance Risk Premium returns.

4.1. Quantified IV curves

Table 3 reports descriptive statistics of the Implied Volatility (IV) factors for SPX options interpolated to six constant maturities across the 2000 to 2025 sample period.

The level factor (γ_0) exhibits an overall increase in its central tendency with increase in maturity, given mean monotonically rising from 0.1779 to 0.1961 and median from 0.1583 to 0.1877 between 1 month to 12 months to maturity. This suggests the At-the-money Implied Volatility (ATM-IV) term structure is

upward-sloping, consistent with contango as the predominant regime over 2000-2025 periods. Over the same period, measures of dispersion and tail behaviour decline with increase in maturity given standard deviation falling from 0.0785 to 0.0487, skewness from 2.2310 to 1.4329, excess kurtosis from 8.1683 to 3.4994 and interquartile range from 0.0857 to 0.0616. This indicates that long-dated IV is smoother and less reactive to spot-volatility shocks as evident from more stable and concentrated distribution, directly reflecting mean reversion. Furthermore, the overall high positive autocorrelation at 1 lag, 5 lag, and 21 lag demonstrate substantial persistence in the level factor, specifically with persistence generally increasing with maturity particularly over the 21-observation horizon.

The slope factor (γ_1) is negative across all maturities, with mean falling from -0.0376 to -0.0492 and median from -0.0334 to -0.0438 between 1 and 12 months. This contradicts the argument that risk-neutral skewness decays with increase in horizons and that the implied volatility SMILE flattens over longer horizons with the likely cause from the standardised moneyness parameterisation. Given the denominator ($\sigma\sqrt{\tau}$) grows with maturity, a given log strike price over forward price maps to a smaller value of said strike price at longer maturities. Additionally, the slope factor represents IV slope per unit standardised moneyness rather than the unit of raw moneyness as mentioned in Zhang and Xiang (2008). Standard deviation rises from 0.0185 at 1 month to 0.0234 at 12-month horizon, with interquartile range rising from 0.0180 to 0.0224 over the

same period indicating dispersion increases with maturity. Excess kurtosis results indicate occurrence of noticeable outliers across all maturities demonstrating heavy tail behaviour and overall, negatively skewed results. Finally, the high positive autocorrelation coefficients indicate strong time varying persistence with a slow decay from an average of 0.953 at 1 lag to 0.622 at 21 lag observation horizons suggesting a potentially greater source of cross-sectional variation in the variance risk premia across regimes.

The curvature factor (γ_2) non-linearly declines with increase in maturity from a mean value of 0.0021 at 1 month to 0.0001 at 12 months to maturity. The interquartile range declines from 0.0029 at 1 month to 0.0016 at 12 months indicating the spread between values generally becomes narrower with increasing maturity. Standard deviation increased from 0.0028 to 0.0069 suggesting the curvature factor experiences increasingly large deviations from its mean at longer maturities. This divergence between declining mean and increasing dispersion indicates that longer dated maturity curvature estimates, while centred near zero, are subject to larger discontinuous deviations. The relationship is likely driven by sparse trading days within the sample period in which the quadratic fit relies on fewer available out-of-the-money strikes, consistent with the declining strike counts documented in Table 1. Skewness varies dramatically across different maturities with a maximum skewness of 11.4197 at 2 months to maturity and minimum skewness of -4.1116 at 3 months to maturity, with an overall absence

of a stable asymmetry across maturities. The large values of excess kurtosis indicate extremely heavy-tailed distribution suggesting curvature is more susceptible to extreme events than the level and slope factors.

Collectively, the three implied volatility curve factors characterise the SPX smirk across the full 2000-2025 sample period as structurally asymmetric, persistently time-varying, and dependent on maturities of each contract in ways that directly motivate the regime-conditional regression framework developed in this study. The level factor is upward sloping, with declining dispersion and strengthening persistence with maturity confirms the intuition that investors' expectations of volatility for long-dated options tends to remain at a stable, long-term average than short-dated options which react sharply and transiently to instantaneous shocks. The slope factor is persistently negative on average across all maturities confirming the markets' chronic demand for downside protection is a structural rather than situational feature of the risk-neutral distribution reflected by the negatively skewed values. The curvature factor, while positive on average, reflects a leptokurtic distribution with excess kurtosis substantially exceeding zero across all maturities indicating considerably heavier tails than a normal distribution reflecting sparse out-of-the-money contracts.

Table 3: Summary of descriptive statistics for Implied Volatility curve factors for SPX options.

Maturity	Factor	Mean	Median	IQR	Std	Skewness	Kurtosis	acf-lag1	acf-lag5	acf-lag21
1m	γ_0	0.1799	0.1583	0.0857	0.0785	2.231	8.1683	0.9766	0.9215	0.7554
1m	γ_1	-0.0376	-0.0334	0.018	0.0185	-2.437	12.3397	0.9367	0.8005	0.5809
1m	γ_2	0.0021	0.0014	0.0029	0.0028	-1.5008	71.6398	0.8322	0.6131	0.5465
2m	γ_0	0.1841	0.1663	0.0838	0.0708	1.9668	6.3895	0.984	0.9432	0.8132
2m	γ_1	-0.0427	-0.0379	0.0199	0.0214	-3.3844	22.9937	0.9576	0.8487	0.6079
2m	γ_2	0.0013	0.0005	0.0021	0.0043	11.4197	227.9626	0.9025	0.6038	0.24
3m	γ_0	0.1867	0.1709	0.081	0.0657	1.8298	5.4968	0.9872	0.9516	0.8378
3m	γ_1	-0.0447	-0.0402	0.0207	0.0209	-2.7567	15.358	0.9556	0.859	0.6336
3m	γ_2	0.0007	0.0002	0.0017	0.0039	-4.1116	361.2449	0.6678	0.4495	0.256
4m	γ_0	0.1889	0.1752	0.0768	0.062	1.7736	5.1751	0.9886	0.9565	0.8526
4m	γ_1	-0.0457	-0.041	0.0214	0.022	-3.1523	19.3891	0.9578	0.8615	0.6515
4m	γ_2	0.0009	0.0001	0.0017	0.0046	6.9609	122.3441	0.8739	0.6825	0.198
6m	γ_0	0.1923	0.1812	0.0715	0.0566	1.6152	4.337	0.9902	0.9624	0.8726
6m	γ_1	-0.047	-0.0423	0.0216	0.0229	-3.4677	22.689	0.9506	0.8568	0.627
6m	γ_2	0.0008	0	0.0014	0.0058	5.497	125.0399	0.7587	0.6279	0.3789
12m	γ_0	0.1961	0.1877	0.0616	0.0487	1.4329	3.4994	0.9933	0.973	0.9055
12m	γ_1	-0.0492	-0.0438	0.0224	0.0234	-3.2306	20.6621	0.9572	0.8572	0.6303
12m	γ_2	0.0001	-0.0004	0.0016	0.0069	2.6159	64.9814	0.8091	0.4772	0.2291

Table 3 summarises the results of the IV curve factors, where γ_0 represents the level, γ_1 represents the slope, and γ_2 represents the curvature of the SPX implied volatility curves. The mean, median, Interquartile range, standard deviation, skewness, kurtosis, autocorrelation at 1 lag, 5 lags, and 21 lags are reported categorised by maturity for each fact

Figure 1 Standard and Poor's Implied volatility quadratic surface fit.

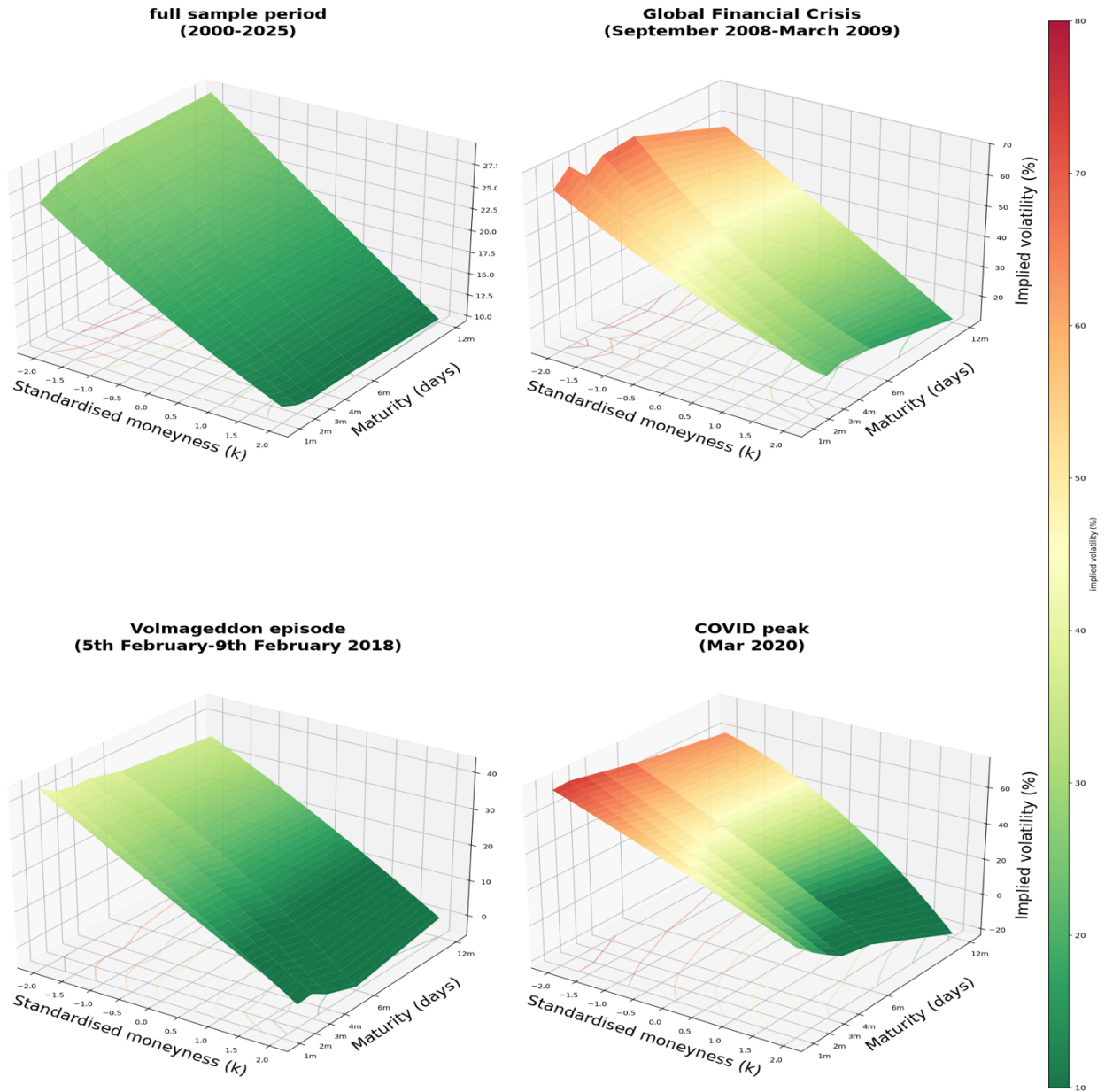


FIGURE 1: S&P 500 implied volatility quadratic surface fit. This figure provides a visual cross-sectional shape of the IV surface across strikes and maturities over the whole sample period (2000-2025), the Global Financial Crisis (GFC: September 2008 – March 2009), the Volmageddon episode (5th February – 9th February), and the COVID-19 pandemic peak (March 2020). Each panel presents the reconstructed IV surface through the level, slope and curvature factors under constant maturity interpolation at 30, 60, 90, 120, and 180 days to maturity. The standardised moneyness (k) on the x-axis, implied volatility (%) on the y-axis and maturity (days) on the z-axis. A colour scale denotes the IV in percentage terms ranging from 10% (green) to 80% (red).

Figure 1 presents the fitted constant maturity implied volatility surface to visualise how the level, slope, and curvature factors interact jointly across the term structure the full sample, the Global Financial Crisis, the Volmageddon episode, and the COVID-19 volatility peak. The full sample surface exhibits a relatively smooth and gently tilted plane with the implied volatility ranging from approximately 10% to 27.5%, as the average over 25 years of data from 2000 to 2025. This consequently implies that calm regimes substantially outnumber crisis episodes confirming that contango is the average state over the sample period. Geometrically, the surface displays a mild negative slope across the standardised moneyness axis at each maturity with options at lower standardised moneyness having relatively higher Implied volatility (IV) than those at higher standardised moneyness. This is in line with the standard SPX equity skew whereby the market assigns higher IV to options providing protection against downside risk relative to those with comparable upside exposure reflecting hedging demand for assets.

The global financial crisis surface deviates substantially from full sample plane in both elevation and shape illustrating that the full sample period acts as a reference plane for average market conditions and obscures conditional regime-specific dynamics. Implied volatility rises to approximately 70% with the peaks more concentrated at shorter maturity with strongly negative standardised moneyness, before declining at longer maturities at less negative moneyness. From this elevated corner, the surface falls steeply in two directions namely,

toward less negative and positive standardised moneyness, consistent with demand for downside protection and crash-risk pricing (Li and Zinna, 2018), and toward longer maturities as it slopes downward representing an inverse relationship of contango in the sample period toward backwardation. This pattern is consistent with the mean-reverting dynamics documented by Mixon (2007), whereby short-dated IV reacts to current prevailing conditions while long-dated IV remains comparatively stable consistent with the assumption that it will later converge towards the long-end. A similar mean-reverting dynamic is present in the COVID-19 surface with similar concentration of IV at short-dated maturity options than long dated options. In comparison to the two crisis periods, the Volmageddon episode is notable modest in absolute elevation with IV peaking at approximately 40%, concentrated at short-dated options, compared to the 70 – 80 % range observed during the global financial crisis and COVID-19. Though, according to Augustin et al. (2021), this period is associated with the abrupt market crash of short volatility strategies on 5th February 2018. As the IV surface represents average constant maturity factors whereby we linearly interpolate maturities closest to set bracket maturities, the spike in realised and implied volatility was concentrated within a single trading session before partially mean-reverting resulting in the five-day period attenuating the peak-day distortion. Overall, the cross-regime comparison shows that elevation, skew, and term structure are regime-dependent features whereby the full sample average conceals rather than summarises these interactions.

4.2. Constant maturity IV curve dynamics

Table 4 reports descriptive statistics for the nine term spread predictors across level, slope and curvature factors. The level spreads are positive on average ranging from 0.016 over the full-term, 0.004 over the long-end, and 0.012 over the short-end spread confirming the upward-sloping ATM Implied volatility term structure documented in Table 3. Backwardation, defined as a negative level spread, occurs on average 21.78% to 26.97% of the sample days between the short-end and long-end spreads respectively. The long-end spread inverts most frequently towards backwardation regardless of absolute spread clustered towards the average with a standard deviation of 0.011 suggesting its more susceptible to volatility oscillations that push it towards zero without jump shocks. All three level spreads exhibit strong negative skewness reflecting an asymmetric distribution that implies that backwardation episodes are more extreme in magnitude than contango episodes despite its rarity. The one period autocorrelation ranges from 0.944 to 0.964, confirming the persistent slow-moving characteristic of the term structure's regime thus justifying the use of the Newey-West standard errors in the predictive regressions.

The slope spreads are negative on average across all maturity pairs ranging from -0.01 at the short end to -0.002 at the long end reflecting the steepening of the volatility skew with increase in maturity in Table 3. Additionally, the slope

spreads exhibit the highest percentage of backwardation over the sample days with 90.01% and 89.73% for the full-term and short end spread respectively. This confirms that the volatility skew is a structure feature of the SPX term structure rather than dependent on distinct regimes. The curvature spreads are similarly negative on average ranging from -0.001, -0.002 and -0.001 for the long, full, and short-term spreads respectively. The excess kurtosis reaches a maximum of 33.529 for the long-end curvature spread consistent with the episode-dependent instability if the curve factor documented in Table 3. Therefore, to predict the VRP results, we will consider the level term and hump spreads with the slope and curvature term structure dynamics serving as continuous predictors, as it provides a genuine and economically distinct relationship with regime conditions.

Table 4 Summary statistics of term spreads.

spread	mean	Std	Min	Max	Skew.	Kurt.	AR (1)	% B
γ_0^{180-30}	0.012	0.029	-0.146	0.055	-2.196	7.442	0.944	21.78
γ_0^{360-30}	0.016	0.038	-0.199	0.068	-2.281	7.862	0.954	22.42
$\gamma_0^{360-180}$	0.004	0.011	-0.048	0.024	-1.522	3.753	0.964	26.97
γ_1^{180-30}	-0.01	0.009	-0.047	0.009	-0.972	2.191	0.851	89.73
γ_1^{360-30}	-0.012	0.011	-0.072	0.015	-1.421	5.419	0.893	90.01
$\gamma_1^{360-180}$	-0.002	0.007	-0.048	0.029	-1.162	10.763	0.806	66.93
γ_2^{180-30}	-0.001	0.003	-0.011	0.018	2.426	14.811	0.8	84.33
γ_2^{360-30}	-0.002	0.005	-0.023	0.034	2.562	25.645	0.803	87.31
$\gamma_2^{360-180}$	-0.001	0.005	-0.042	0.033	-1.914	33.519	0.739	65.22

Table 4 summarises descriptive statistics for the term spread predictors described in section 3.3. for the level, slope, and curvature factors computed across the short-end (180-30), long-end (360-180), and full term spread (360-30) days to maturity. For each spread, the table reports the mean, standard deviation, minimum, maximum, skewness, kurtosis, first order autocorrelation, and the percentage of days when the spread was negative (% B), corresponding to the backwardation regime.

Figure 2 presents the 21-day rolling averages of the ATM IV term spreads over short-end representing 180 to 30 days, the long-end representing 360 to 180 days, and the full-term spread representing 360 to 30 days to maturity. Across the three panels, contango represented by the blue shading is the visually dominant regime outside of known crisis windows with spreads oscillating within narrow positive bands of 0.00 to 0.05 for short-end spreads, 0.00 to approximately 0.052 for full-term spreads, and 0.00 to 0.03 for long-end spreads. This is consistent with the positive means reported in Table 4. Shallow backwardation regimes occur across all panels regardless of being situated within calm periods, especially for the long-end spreads, consistent with findings that the IV term structure does not maintain a strictly monotonic slope and is susceptible to jump risk (Christoffersen et al., 2009; Todorov, 2010).

The full-term spreads exhibit the widest range in both contango and backwardation regimes as they accumulate the combined dynamics of the short and long ends of the implied volatility surface. Specifically, the short-end spreads exhibit a compressed pattern in comparison with the full-term spreads with relatively similar backwardation episodes. In comparison, the long-end exhibits a relatively more muted pattern except for during the COVID-19 and Federal reserve rate hike periods whereby backwardation regimes lasted for a longer period than in the other panels. This confirms that short-dated Implied volatility absorbs most reactions to spontaneous market shocks while long-dated IV acts as

a relative anchor provided the market can accurately predict how long a given short-term crisis will last and whether it will remain persistent long term thus reflecting expectations of convergence toward a long-run volatility level.

The early sample period from 2000 to 2003 produces most severe backwardation cluster ranging from approximately -0.085 over the short-end, -0.03 over the long-end, and -0.12 over the full-term spread. This was likely driven by the sequential combination of the Dot-Com bust, the 9/11 attacks, and the 2002 accounting scandal. The Global Financial Crisis resulted in the deepest trough amongst all spreads whereby it persisted over several months. In comparison the US debt ceiling crisis and Volmageddon episode produced relatively shallower and narrower troughs consistent with being limited episodes without broad systematic repricing events. This justifies the use of a regime-conditional indicator within the regression rather than a single time-invariance coefficient.

Figure 2 The level factor term spreads under a 21-day rolling average.



Figure 3 Time series of level term spreads with regime identification. This figure presents the three At-the-Money Implied Volatility term spread factors defined in section 3.3. as γ_0^{360-30} representing the spread between 360 to 30 days to maturity, $\gamma_0^{360-180}$ representing the spread between 360 to 180 days to maturity, and γ_0^{180-30} representing the spread between 180 to 30 days to maturity. These are computed as 21-day rolling averages over the sample period with shaded regions distinguishing contango (spread > 0, in light blue) and backwardation (spread < 0, in red). Vertical markers indicate known volatility episodes namely: GFC – global financial crisis in 2008, the US debt ceiling crash in 2011, the Volmageddon episode in 2018, the Federal Reserve rate hike in 2022, the COVID-19 pandemic in 2020, and President Donald Trump’s Liberation Day tariff announcement in 2025.

4.3. Variance risk premium results

Table 5 reports the descriptive statistics for the daily synthetic variance swap and delta-hedged straddle returns across constant maturities up to 1 month, 2 months, 3 months, 4 months, 6 months and 12 months over the sample period.

The ATM straddle assets produce negative mean returns across all maturities from 1 month to 6 months ranging from - 0.328 % at 1 month to -0.052% at 6 months to maturity per day, before later turning positive at 12 months to maturity with 0.02 % daily mean returns. This monotonic increase in daily mean returns is consistent with the maturity-depended assumptions of the variance risk premium in which buyers of short-dated ATM straddles overpay relative to realised volatility to achieve short-term downside protection. The mean returns compress towards zero as maturity increases as implied volatility converges towards long-run unconditional levels as documented in Figure 1 during crisis periods. Similarly, the standard deviation of returns declines monotonically from 5.513% to 1.165% indicating more volatility in returns for shorter maturity assets than longer maturity assets, consistent with the dampening of underlying asset price volatility sensitivity at greater maturities. Returns are positively skewed with a leptokurtic distribution of returns indicate returns are clustered at shorter maturities. The first-order autocorrelation rises with maturity indicating daily straddle returns exhibit minimal serial dependence with the risk premium rapidly incorporating option prices.

The synthetic variance swap returns produce positive daily mean returns ranging from 5.857% at 1 month to 0.057% at 6 months to maturity, with 12 months to maturity turning negative with mean returns of -0.141% aligning in sign with straddle returns. This suggests a primary divergence in returns between the two assets across most maturities with variance swap returns not aligning with the mechanisms of variance risk premium returns. This is likely attributed to near-zero strip value amplification within construction of variance swap returns. Given variance swap return is computed as the weighted sum of all out-of-the-money (OTM) put and call mid prices over the next day as the strip value divided by the same weighted sum minus one. During extended low volatility regimes, deep OTM options trade at near-zero prices thus driving the strip value towards zero. Therefore, during the revaluing function within construction of variance swap returns, extreme percentage returns are generated relative to the near zero-denominator the previous day. This could be attributed to the translation of the variance swap return construction methodology into code whereby errors were made leading to returns that are inconsistent with the theoretical interpretation of the asset as variance risk premium proxies.

Given this limitation, the variance swap returns within the regression analysis are to be interpreted with caution. The straddle returns, which do not exhibit this amplification effect, therefore serve as the primary variance risk premium proxy.

Table 5 Summary of VRP descriptive statistics on daily returns.

maturity	VRP asset	Mean (%)	std (%)	Skew.	Kurt.	Min (%)	max (%)	AR (1)
1m	variance swap	5.857	20.245	2.656	9.782	-24.436	124.903	0.1
1m	straddle	-0.328	5.513	1.886	5.922	-11.254	27.776	-0.023
2m	variance swap	2.328	12.798	2.571	10.256	-20.664	80.899	0.047
2m	straddle	-0.199	3.342	1.692	5.749	-7.66	17.362	0.018
3m	variance swap	1.022	9.428	2.299	8.547	-16.449	55.926	0.027
3m	straddle	-0.114	2.605	1.476	5.023	-6.628	13.089	0.048
4m	variance swap	0.378	7.699	2.021	7.218	-15.346	43.844	0.016
4m	straddle	-0.083	2.154	1.327	4.647	-5.775	10.685	0.055
6m	variance swap	0.057	6.013	1.703	5.668	-13.773	31.473	-0.008
6m	straddle	-0.052	1.713	1.187	4.474	-4.796	8.527	0.056
12m	variance swap	-0.141	3.794	1.158	3.72	-10.301	17.043	-0.003
12m	straddle	0.02	1.165	0.894	3.115	-3.195	5.267	0.104

Table 5 summaries descriptive statistics of the variance risk premium (VRP) assets' daily returns described in section 3.4. These assets include daily synthetic variance swaps and ATM straddle returns across six constant maturities over the full 2000-2025 sample period. The descriptive statistics include mean, standard deviation, skewness, kurtosis, minimum, maximum and one period autocorrelation lag of daily returns for each asset.

4.4. In-sample predictability results

Table 6 presents the in-sample regression results for the regime-conditional predictive regression for monthly straddle (a) and synthetic variance swap (b) returns on the three level factor term spreads. We applied the Newey-West heteroscedasticity and autocorrelation statistic with 4 lags for weekly and 20 lags for monthly returns under 1-month, 2,3,4,6 and 12-months to maturity. Appendix T2 indicates that the monthly horizon returns provide the overall strongest t-statistics at medium-to-long maturities. Table 6 (b) shows that 1-month to 3 months to maturity variance swap regression results produced no valid monthly observations and were excluded from the inference due to the revaluation transition resulting in invalid results as mentioned in section 4.3. The Newey-West T-statistic for the coefficients of the regression analysis suggest spreads at 4 month and 6 months to maturity are statistically significant with p-values less than 0.05 (γ_0^{360-30} and $\gamma_0^{360-180}$ at 4 months, γ_0^{180-30} at 6 months). Given observation counts were 100, 124, and 403 respectively, their interpretations should be taken with caution. Table 6 (a) shows that the contango slope (β_1) was negative and statistically significant at 3 months ($\beta_1 = -1.555, p = 0.017$), 4 months ($\beta_1 = -1.397, p = 0.018$), and 6 months ($\beta_1 = -2.775, p = 0.006$). Specifically, the short end (γ_0^{180-30}) spread resulted as the most consistent significant predictor of straddle returns. This is consistent with the assumption that during contango, defined as the market pricing in calm conditions in the

present but expects more uncertainty in the future resulting in a positive term spread, a wider-positive term spread predicts more negative subsequent monthly straddle returns at medium maturities. The full-term spread shows marginal significance, when extending the p-value to 0.1, at 4 months ($p = 0.080$) and 6 months ($p = 0.055$) to maturity while the long end spread results in uniformly insignificant predictive power at all maturities.

The regime shift coefficient (β_3) was largely insignificant for 1 month to 6 months to maturity indicating that the predictive relationship is regime – stable, whereby the relationship did not shift in strength, and the backwardation slope coefficient did not change in sign. The notable exception was at 12 months to maturity, specifically the full-term and long-term spreads whereby the regime shift coefficient was significant whereby $\beta_3 = -1.443, p = 0.040$ and $\beta_3 = -5.568, p = 0.012$ respectively. As the regime indicator is $D_t = 0$ during contango and $D_t = 1$ during backwardation, for the long-end spread at 12-months to maturity, the contango slope is positive (+1.894) while the backwardation slope is negative (-3.674). Similarly, for the full-term spread at 12-months to maturity, the same sign flip was noticed with the contango slope as positive (+0.446) and backwardation slope as negative (-0.995). This sign flip, specifically with a greater difference in the long end, can be interpreted as the level term spread signalling that the structural long-run uncertainty during contango is priced into straddle returns resulting in buyers of long-dated straddles experiencing relatively

smaller losses, while in backwardation the same spread signals that the market has overpriced long-dated variance with subsequent mean reversion results in significantly greater losses for the same position resulting in a negative premium. This indicates that the term spread can be interpreted differently depending on the regime and presents evidence that the regime indicator provides additional information at 12 months to maturity to distinguish between structural and transient variance dynamics.

Table 7 presents the in-sample regression results for the regime-conditional regression for monthly straddle (a) and monthly variance swap (b) the hump factors for the level (H_0), slope (H_1), and curvature (H_2) estimated over the 2000 to 2025 in-sample regression period.

Similar to the level spread results in Table 6 (b), the 1 to 3 months to maturity variance swap maturities produced no valid observations. At 4 month and 6-month, results show extremely large coefficient magnitudes ranging from -1602.2287 to -658.8199 likely driven by the limited number of observations. Table 7 (a) shows that the level hump factor (H_0) was the dominant predictor of VRP returns with statistically significant coefficients (β_1) at 2 months ($p = 0.010$), 3 months ($p = 0.010$), 6 months ($p = 0.016$), and 12 months ($p = 0.040$) to maturity. As the β_1 coefficient is positive when the level hump factor is significant at 2 months, 3 months, and 12 months to maturity, this suggest that

level hump factor more negative ($H_0 < 0$) therefore indicating that the curve bulges upward in the middle and flattens toward the long end. This is consistent with the elevated medium-term VRP shown in Table 5 associated with buyers of variance overpaying. Therefore, the hump factor captures additional cross-maturity curvature information beyond what the level term spread captures, specifically whether the middle of the implied volatility term structure is independently elevated beyond its long term (360-days) and short term (30-days) would predict. As β_3 coefficient is largely insignificant from maturities at 2 months to 12 months, this shows that the relationship between the VRP returns and the hump factor is more regime-stable than the full-term spreads with an exception at the 1 month to maturity level hump factor.

Table 6 (a) In-sample level term spread straddle regression.

Maturity	Predictor	α	β_1	$\beta_1 - T$	$\beta_1 - p$	β_2	β_3	$\beta_3 - T$	$\beta_3 - p$	Contango slope	Backwardation slope	R_{IS}^2	N
1m	γ_0^{180-30}	-0.100	0.355	0.187	0.852	-0.018	0.550	0.260	0.795	0.355	0.905	0.010	813
1m	γ_0^{360-30}	-0.131	1.186	0.873	0.383	0.002	-0.755	-0.452	0.651	1.186	0.432	0.012	813
1m	$\gamma_0^{360-180}$	-0.142	5.021	1.103	0.270	-0.032	-5.955	-0.820	0.412	5.021	-0.934	0.020	813
2m	γ_0^{180-30}	-0.038	-1.279	-1.633	0.103	-0.045	-0.995	-0.807	0.420	-1.279	-2.274	0.063	2834
2m	γ_0^{360-30}	-0.051	-0.445	-0.729	0.466	-0.026	-1.116	-1.177	0.239	-0.445	-1.561	0.051	2834
2m	$\gamma_0^{360-180}$	-0.067	1.347	0.639	0.523	0.006	-5.216	-1.522	0.128	1.347	-3.869	0.022	3231
3m	γ_0^{180-30}	-0.017	-1.555	-2.378	0.017	-0.038	0.141	0.150	0.881	-1.555	-1.415	0.063	2214
3m	γ_0^{360-30}	-0.026	-0.760	-1.465	0.143	-0.027	-0.262	-0.358	0.720	-0.760	-1.023	0.051	2214
3m	$\gamma_0^{360-180}$	-0.046	-0.019	-0.012	0.991	-0.006	-3.242	-1.345	0.179	-0.019	-3.261	0.030	2490
4m	γ_0^{180-30}	-0.010	-1.397	-2.367	0.018	-0.028	0.518	0.550	0.582	-1.397	-0.879	0.044	2160
4m	γ_0^{360-30}	-0.014	-0.793	-1.751	0.080	-0.023	0.187	0.260	0.795	-0.793	-0.607	0.033	2160
4m	$\gamma_0^{360-180}$	-0.022	-1.311	-1.055	0.291	-0.019	-0.481	-0.210	0.834	-1.311	-1.792	0.012	2453
6m	γ_0^{180-30}	-0.004	-1.191	-2.775	0.006	-0.028	0.268	0.441	0.659	-1.191	-0.923	0.059	2994
6m	γ_0^{360-30}	-0.009	-0.650	-1.922	0.055	-0.024	-0.045	-0.095	0.924	-0.650	-0.695	0.050	2994
6m	$\gamma_0^{360-180}$	-0.021	-0.512	-0.471	0.638	-0.016	-2.222	-1.370	0.171	-0.512	-2.734	0.038	3414
12m	γ_0^{180-30}	-0.013	0.313	0.446	0.656	-0.033	-1.612	-1.729	0.084	0.313	-1.299	0.073	1403
12m	γ_0^{360-30}	-0.017	0.446	0.864	0.387	-0.027	-1.443	-2.055	0.040	0.446	-0.998	0.076	1403
12m	$\gamma_0^{360-180}$	-0.019	1.894	1.396	0.163	-0.021	-5.568	-2.501	0.012	1.894	-3.674	0.070	1646

Table 6 (a) summarises in sample regression analysis results for the level term spread straddle regression.

Table 6 (b) In-sample level term spread synthetic variance swap regression.

Maturity	Predictor	α	β_1	$\beta_1 - T$	$\beta_1 - p$	β_2	β_3	$\beta_3 - T$	$\beta_3 - p$	Contango slope	Backwardat ion slope	R_{IS}^2	N
3m	γ_0^{180-30}												
3m	γ_0^{360-30}												
3m	$\gamma_0^{360-180}$												
4m	γ_0^{180-30}	0.0412	14.0699	1.764	0.0777	-0.1305	-12.805	-1.524	0.1275	14.0699	1.2649	0.2454	100
4m	γ_0^{360-30}	-0.0344	21.23	3.099	0.0019	-0.0661	-20.511	-3.18	0.0015	21.23	0.7191	0.2037	100
4m	$\gamma_0^{360-180}$	0.0349	-35.6136	-2.798	0.0051	-0.143	35.4725	2.875	0.004	-35.6136	-0.1411	0.0334	124
6m	γ_0^{180-30}	0.1885	-7.3609	-2.727	0.0064	-0.2606	7.4049	2.459	0.0139	-7.3609	0.044	0.133	403
6m	γ_0^{360-30}	0.1353	-1.7647	-0.49	0.6244	-0.2262	1.4818	0.411	0.6809	-1.7647	-0.2829	0.1363	403
6m	$\gamma_0^{360-180}$	-0.0153	27.1559	1.059	0.2894	-0.0944	-30.5058	-1.182	0.2374	27.1559	-3.3499	0.1653	479
12m	γ_0^{180-30}	-0.0517	-0.1593	-0.128	0.8981	0.0281	0.3955	0.244	0.8072	-0.1593	0.2362	0.0088	580
12m	γ_0^{360-30}	-0.0252	-1.4761	-1.226	0.2202	-0.0089	1.4897	1.034	0.3012	-1.4761	0.0136	0.0114	580
12m	$\gamma_0^{360-180}$	-0.0064	-7.4545	-1.534	0.125	-0.0435	5.4328	0.932	0.3515	-7.4545	-2.0217	0.0247	696

Table 6 (b) summarises in-sample regression analysis results for the level term spread synthetic variance swap regression.

Table 6 (a) and (b) presents the in-sample regression results for the regime-conditional predictive regression of monthly straddle and synthetic variance swap returns respectively for three level factor term spreads namely: γ_0^{180-30} representing the short end spread, γ_0^{360-30} representing the full term spread and $\gamma_0^{360-180}$ representing the long end spread. The table reports α as the intercept, the contango slope β_3 , its Newey-West t-statistic at 20 lags as $\beta_3 - T$ and p-value as $\beta_3 - p$, the effective backwardation slope $\beta_1 + \beta_3$, the in-sample coefficient of determination R_{IS}^2 , and the number of valid observations as N at 1 month, 2 months, 3 months, 4 months, 6 months and 12 months to maturity over the sample period from 2000 to 2025.

Table 7 (a) In-sample monthly hump factors for straddle returns

Maturity	Hump	α	β_1	$\beta_1 - T$	$\beta_1 - p$	β_2	β_3	$\beta_3 - T$	$\beta_3 - p$	Contango slope	Backwardat ion slope	R^2_{IS}	N
1m	H_0	-0.1327	-190.0039	-1.145	0.2521	0.1019	730.0686	1.997	0.0458	-190.0039	540.0646	0.0255	813
1m	H_1	-0.0693	-534.8786	-1.897	0.0578	-0.0078	-69.3926	-0.083	0.9342	-534.8786	-604.2712	0.0207	813
1m	H_2	-0.1232	796.2559	1.786	0.0741	0.0909	535.2831	1.085	0.2779	796.2559	1331.539	0.0355	813
2m	H_0	-0.0823	478.1066	2.58	0.0099	0.0534	-87.4734	-0.368	0.713	478.1066	390.6332	0.0779	2834
2m	H_1	-0.0407	-80.3239	-0.363	0.7164	-0.036	-432.8472	-0.764	0.4449	-80.3239	-513.171	0.0026	2834
2m	H_2	-0.0684	539.3192	1.45	0.147	0.0492	-557.6888	-0.745	0.4563	539.3192	-18.3697	0.0109	2834
3m	H_0	-0.0526	281.9579	2.564	0.0103	0.0449	157.1194	0.905	0.3657	281.9579	439.0773	0.0782	2214
3m	H_1	-0.0269	-176.9859	-1.169	0.2423	-0.0162	-223.7834	-0.862	0.3887	-176.9859	-400.7694	0.0095	2214
3m	H_2	-0.055	436.3077	1.675	0.0939	0.0407	-532.9017	-1.255	0.2094	436.3077	-96.5941	0.0181	2214
4m	H_0	-0.0387	189.7273	1.501	0.1332	0.032	168.2858	0.952	0.341	189.7273	358.0131	0.0603	2160
4m	H_1	-0.014	-259.3738	-2.952	0.0032	-0.0225	-71.3226	-0.286	0.7749	-259.3738	-330.6964	0.0214	2160
4m	H_2	-0.0363	95.7529	0.299	0.765	0.0231	-29.1614	-0.065	0.9478	95.7529	66.5915	0.0052	2160
6m	H_0	-0.0298	174.6165	2.42	0.0155	0.0284	125.5968	1.122	0.2618	174.6165	300.2133	0.0681	2994
6m	H_1	-0.0179	-46.5793	-0.498	0.6188	-0.0135	-177.7858	-0.961	0.3365	-46.5793	-224.3652	0.0024	2994
6m	H_2	-0.0316	348.6056	1.643	0.1005	0.0133	-471.1071	-1.588	0.1122	348.6056	-122.5015	0.0119	2994
12m	H_0	-0.0383	217.5445	2.052	0.0401	0.0326	-189.9245	-1.096	0.2733	217.5445	27.62	0.0562	1403
12m	H_1	-0.0055	-2.9922	-0.023	0.9816	-0.0101	133.4939	0.721	0.471	-2.9922	130.5017	0.0075	1403
12m	H_2	-0.008	164.6509	0.668	0.5039	-0.0135	-143.4245	-0.377	0.7061	164.6509	21.2264	0.012	1403

Table 7(a) summarises in-sample regression analysis results for the level, slope, and curvature hump factors for the straddle returns regression.

Table 7(b) In-sample hump factors for synthetic variance swaps.

Maturity	Hump	α	β_1	$\beta_1 - T$	$\beta_1 - p$	β_2	β_3	$\beta_3 - T$	$\beta_3 - p$	Contango slope	Backwardation slope	R^2_{IS}	N
3m	H_0												
3m	H_1												
3m	H_2												
4m	H_0	-0.0673	-359.1271	-1.502	0.133	0.2666	759.7905	1.385	0.166	-359.1271	400.6634	0.2885	100
4m	H_1	-0.0101	-1602.2287	-1.686	0.0918	-0.0544	3055.7777	2.16	0.0308	-1602.2287	1453.5491	0.1219	100
4m	H_2	-0.046	-5.3857	-0.018	0.9859	-0.1063	440.8111	0.626	0.531	-5.3857	435.4255	0.1281	100
6m	H_0	-0.0453	-120.8998	-0.724	0.4688	0.2382	1727.4444	1.54	0.1236	-120.8998	1606.5445	0.117	403
6m	H_1	0.0357	-658.8199	-1.461	0.1441	-0.0789	1249.3071	1.722	0.0851	-658.8199	590.4872	0.031	403
6m	H_2	-0.0303	679.144	1.106	0.2689	-0.0158	-429.8235	-0.62	0.5353	679.144	249.3205	0.0165	403
12m	H_0	-0.0185	-68.5046	-0.507	0.6124	-0.0632	-252.6728	-0.866	0.3862	-68.5046	-321.1774	0.022	580
12m	H_1	-0.0216	-380.7894	-1.131	0.2582	-0.0209	434.5962	0.986	0.3241	-380.7894	53.8068	0.0092	580
12m	H_2	-0.044	292.6072	0.84	0.4011	0.0104	275.8674	0.52	0.6027	292.6072	568.4747	0.0194	580

Table 7 (b) summarises in sample regression analysis results for the level, slope, and curvature hump factors for the straddle returns regression.

Table 7 (a) and (b) presents the in-sample regression results for the regime-conditional predictive regression monthly straddle and synthetic variance swap returns respectively for the level, slope, and curvature hump factors namely: H_0 representing the level hump, H_1 representing the slope hump and H_2 representing the curvature hump.

4.5. Out-of-sample results

Table 8 presents the out-of-sample evaluation metrics for the monthly straddle returns against the level term factors. Appendix T3 indicated the model produced negative coefficient of determination (R_{os}^2), reflecting limitations mentioned in Section 4.3.

The directional accuracy was significant only for 1 month and 3 months to maturity results except for short-end spread at 2 months despite the negative R_{os}^2 . The long end spread level factor ($\gamma_0^{360-180}$) at 12 months to maturity achieves a R_{os}^2 of 0.0512 with most significant p-value ($p < 0.01$) within the out-of-sample regression at a Clark-West T-statistic of 2.707 ($p = 0.0034$). The monotone improvement in performance out-of-sample results from the short end to the long-end spread at 12 months to maturity indicates that the regression model effectively captures the structural behaviour of the market in which the 360-to-180-day ATM Implied Volatility spread directly prices the long-run risk premium relevant to 12-months to maturity straddle portfolios. Additionally, it shows that the instability of the short-end spread out of sample results could be attributed to being more sensitive to the regime-switching dynamics. This was likely due to the in-sample regression data (2000 to 2015) accounting for more pronounced crisis events, namely the Great Financial Crisis and the Dot-Com bubble, which

led to the model attempting to generalises the likelihood of these spikes into the structurally different volatility crises within the out-of-sample period from 2016 to 2025, as illustrated in Figure 3.

Table 9 presents the Out-of-sample metrics for the monthly hump factors. Despite the level, slope and curvature hump factors resulting in significant directional accuracy from 1 month to 3 months to maturity options, the R^2_{os} produced negative results through the same period. The only positive coefficient of determination was for the 4 months to maturity curvature hump with marginal significance ($p < 0.1$) of 0.0992 indicating that the historical mean benchmark consistently outperformed most of the hump factor models. The most significant underperformance of the model was at 12 months to maturity slope and curvature factors with Clark-West p-values of 0.9799 and 0.9925 respectively. This was likely similar to the in-sample overfitting to crisis dynamics within the term spreads out-of sample results. Overall, the hump factors contribute to directional predictive information for out-of-sample results at short maturities but fail to generalise over most of the 2016 to 2025 period.

Table 8 Out of sample level term factors for straddle returns.

Maturity	Predictor	R_{OS}^2	CW stat	CW p-value	DA	DA p-value	No. OOS
1m	γ_0^{180-30}	-0.0067	-0.553	0.7099	0.7049	0.0009	1267
1m	γ_0^{360-30}	-0.0069	-0.345	0.6349	0.7049	0.0009	1267
1m	$\gamma_0^{360-180}$	-0.0057	0.164	0.4348	0.7049	0.0009	1267
2m	γ_0^{180-30}	-0.046	-0.196	0.5775	0.6604	0.0135	1108
2m	γ_0^{360-30}	-0.0187	0.058	0.4767	0.6415	0.0267	1108
2m	$\gamma_0^{360-180}$	-0.013	0.29	0.386	0.5849	0.1358	1108
3m	γ_0^{180-30}	-0.0104	0.552	0.2906	0.5981	0.0263	2233
3m	γ_0^{360-30}	0.0044	0.68	0.2481	0.5981	0.0263	2233
3m	$\gamma_0^{360-180}$	0.0048	0.812	0.2083	0.5888	0.0407	2233
4m	γ_0^{180-30}	-0.0127	0.467	0.3204	0.57	0.0967	2084
4m	γ_0^{360-30}	0.0066	0.732	0.2322	0.57	0.0967	2084
4m	$\gamma_0^{360-180}$	0.0045	0.675	0.2498	0.55	0.1841	2084
6m	γ_0^{180-30}	0.0154	0.68	0.2484	0.5417	0.3327	993
6m	γ_0^{360-30}	0.0484	0.837	0.2014	0.5417	0.3327	993
6m	$\gamma_0^{360-180}$	0.0652	0.99	0.1611	0.5208	0.4427	993
12m	γ_0^{180-30}	0.0103	1.436	0.0755	0.4487	0.8459	1635
12m	γ_0^{360-30}	0.0218	1.785	0.0372	0.5128	0.455	1635
12m	$\gamma_0^{360-180}$	0.0512	2.707	0.0034	0.5897	0.0703	1635

Table 8 summarises the evaluation metrics for the out of sample regression analysis level factor term spreads, applied over the 2016-2025 period. R_{OS}^2 represents the Campbell-Thompson coefficient of determination, CW stat and CW p-value represent the Clark-West MSPE-statistic and its associated p-value respectively.

Table 9 Out-of-sample hump factors for straddle returns.

Maturity	Predictor	R_{Os}^2	CW stat	CW p-value	DA	DA p-value	No. OOS
1m	H_0	-0.0211	-0.728	0.7668	0.7049	0.0009	1267
1m	H_1	-0.0027	0.671	0.251	0.7049	0.0009	1267
1m	H_2	-0.0065	0.008	0.4966	0.7049	0.0009	1267
2m	H_0	-0.1189	-0.162	0.5644	0.6604	0.0135	1108
2m	H_1	-0.0054	-0.354	0.6382	0.6604	0.0135	1108
2m	H_2	-0.0222	-0.236	0.5934	0.6604	0.0135	1108
3m	H_0	-0.0618	0.395	0.3464	0.5981	0.0263	2233
3m	H_1	-0.0076	-0.394	0.6531	0.6075	0.0165	2233
3m	H_2	-0.0064	0.291	0.3856	0.6168	0.01	2233
4m	H_0	-0.0754	0.078	0.4689	0.58	0.0666	2084
4m	H_1	-0.0148	-0.371	0.6446	0.55	0.1841	2084
4m	H_2	0.0027	1.286	0.0992	0.57	0.0967	2084
6m	H_0	-0.0578	0.442	0.3294	0.5417	0.3327	993
6m	H_1	-0.008	-0.616	0.7311	0.5208	0.4427	993
6m	H_2	0.0403	0.942	0.173	0.5625	0.2354	993
12m	H_0	-0.0177	0.247	0.4024	0.4487	0.8459	1635
12m	H_1	-0.0293	-2.052	0.9799	0.4359	0.8937	1635
12m	H_2	-0.0334	-2.433	0.9925	0.4231	0.9297	1635

Table 9 summarises the evaluation metrics for the out of sample regression analysis level, slope, and curvature hump factors, applied over the 2016 – 2025 sample period.

5. Conclusion

This study examines whether the shape of the Standard and Poor's 500 implied volatility structure (VIX) predicts the variance risk premium (VRP) across maturities and market regimes, extending the framework of Guo et al. (2023) along a substantially greater sample period covering January 2000 to July 2025. This paper introduces a regime-conditional regression framework to investigate the predictive relationship of the term structure to the variance risk premium returns within contango and backwardation regimes. Additionally, the paper implements a rate-normalised hump factor that captures the non-linear medium term curvature dynamics beyond a standard term structure endpoint-based spread. Finally, the paper performs a robustness test using out of sample data to examine the significance of predictors for variance asset returns with a directional accuracy test to consistency of predictors.

Following the framework proposed by Zhang and Xiang (2008), the implied volatility (IV) smirk is decomposed into level (r_0), slope (r_1), and curvature (r_2) factors at six constant maturities namely 30 days, 60 days, 90 days, 120 days, 180 days, and 360 days matching the CBOE VIX-style term structure conventions. The level factor is upward sloping on average, confirming contango as the average state of the VIX index. Its persistence with increase in days to maturity reflect the mean-reverting dynamics of long-run variance expectations as

documented by Mixon (2007). The slope factor is persistently negative across all maturities, confirming that the Standard and Poor's 500 index (SPX) volatility skew is a structure feature of the risk-neutral distribution of market expectations. The curvature factor is near-zero on average whilst exhibiting extreme instability at medium maturities consistent reflected by large kurtosis results. This episodic dependency mimics the VIX smile, whereby options that are far out-of-the money (OTM) exhibits higher implied volatility than At-the-money (ATM) options.

This paper proxies the variance risk premium returns using variance-sensitive assets, following the methodology applied by Coval and Shumway (2001) for straddle portfolio returns and Dew-Becker et al. (2017) for synthetic variance swaps, within the in-sample regression from 2000 to 2015 and out-of-sample regression from 2016 to 2026. The in-sample regressions show that the level factor short-end spread (r_0^{180-30}) predicts monthly straddle returns with statistically significant negative coefficients at 3 months through to 6-month maturities, consistent with buyers of variance overpaying in contango. The regime-conditional indicator added additional information specifically at the 12-month maturity, where the long-end spread ($r_0^{360-180}$) exhibits a sign shift in the predictive slope between contango (+1.894%) and backwardation (-3.674). The hump level factor (H_0) provided supplementary evidence at 2-month to 12-month maturities, confirming the non-linear curvature of the medium term IV surface consistent with the theoretical motivation from Christoffersen et al. (2009) and

Todorov (2010). In the out-of-sample evaluation, the long-end level spread ($r_0^{360-180}$) at 12 months is the study's most robust result, achieving a coefficient of determination of 0.051, with a Clark-West statistic of 2.707 ($p = 0.0034$). In contrast, the hump factors fail to generalise to the out-of-sample data despite strong in-sample significance, as the model carried forward the structurally different volatility dynamics from the estimation window to the out-of-sample data. Regardless, they retain statistically significant directional accuracy at short maturities providing evidence of genuine directional information content.

Comparing these findings against the existing literature, we find that consistent monthly-horizon predictability across all three compounding horizons for straddle returns, while Guo et al. (2023) document dominance at the one-week horizon. This could likely be attributed to additional volatility regime variation within the extended dataset as well as directly developing variance risk premium data rather than sourcing from Johnson. (2017) whose dataset ended in 2017. Additionally, the out-of-sample results identify the long-end spread as the most robust predictor of VRP returns rather than short-end spreads complementing Guo et al. (2023)'s emphasis on near term structural information.

6. Limitations.

First, the binary regime indicator, $D_t \in \{0,1\}$ where $D_t = 0$ denotes positive term spread and $D_t = 1$ denotes negative term spreads, discards the magnitude of market stress within each regime. This reduces its information capacity within the regression analysis to merely a direction indicator that cannot distinguish between mild contango or severe backwardation. A continuous interaction variable that changes with the spread factor would ideally allow the predictive slope to vary more with regime intensity providing additional information.

Second, the synthetic variance swap returns exhibit a near zero strip value during extended calm periods which generated extreme percentage returns during subsequent revaluing events. This limited the study to only rely on straddle returns as the proxy of variance risk premium. A possible solution would be implementing propriety over-the-counter variance swap quotes as in Konstantinidi and Skiadopoulos (2016).

Finally, the hump factor was measured using daily closing option prices, which likely understated the frequency of its formation within intraday option prices. This limited the predictive ability of the hump factor within shorter volatility spike crisis periods such as Volmageddon (Augustin et al., 2021).

7. Directions for future research.

By replacing the binary regime indicator with a logarithmic transition function that varies continuously with the term spread or the hump factor spread, the predictive slope would transition gradually between regimes rather than switching at zero.

Additionally, by extending the analysis to VIX futures returns as a third variance-sensitive asset would provide a more robust and cleaner out of sample test with different independent variable for the regression analysis.

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7.1. APPENDIX T1 Data validation table.

Maturity bucket	Number of observations	Mean difference (%)	Median difference (%)
<30d	36271	0.49%	0.24%
30-90d	36757	0.57%	0.27%
90-180d	25763	0.64%	0.32%
180-360d	25824	0.83%	0.42%
>360d	23579	2.29%	0.80%

Table summarising comparison between implied forward price obtained from put-call parity at minimised strike vs ordinary least squares.

7.2. APPENDIX T2 Horizon comparison table for in-sample regression analysis

Asset	Maturity	β_1 daily	$\beta_1 - T$ daily	R_{IS}^2 daily	β_1 weekly	$\beta_1 - T$ weekly	R_{IS}^2 weekly	β_1 monthly	$\beta_1 - T$ monthly	R_{IS}^2 monthly
variance swap	1m	-1.0083	-1.75	0.0031						
variance swap	2m	-0.0733	-0.28	0.0007	0.7375	0.436	0.0221			
variance swap	3m	-0.0457	-0.277	0.0003	-0.1042	-0.08	0.0158			
variance swap	4m	-0.0681	-0.518	0.0004	-0.0952	-0.081	0.0055	14.069	1.764	0.2454
variance swap	6m	-0.0678	-0.627	0.0017	-0.794	-1.325	0.0034	-7.3609	-2.727	0.133
variance swap	12m	0.0241	0.28	0.0003	0.5117	1.345	0.0104	-0.1593	-0.128	0.0088
straddle	1m	-0.1184	-1.302	0.0019	-0.4154	-1.348	0.0079	0.3554	0.187	0.0095
straddle	2m	-0.1005	-1.957	0.0042	-0.3023	-1.815	0.0311	-1.2788	-1.633	0.0629
straddle	3m	-0.0838	-2.068	0.0061	-0.2987	-2.134	0.0378	-1.5553	-2.378	0.0626
straddle	4m	-0.0746	-2.271	0.0032	-0.254	-2.117	0.0259	-1.3972	-2.367	0.0441
straddle	6m	-0.0572	-2.08	0.005	-0.2276	-2.258	0.0278	-1.1914	-2.775	0.0586
straddle	12m	-0.0361	-1.4	0.0026	-0.0309	-0.291	0.0242	0.313	0.446	0.0725

Table compares in sample predictability of the level factor across daily, weekly, and monthly return horizons for variance swap and straddle returns.

7.3. APPENDIX T3 Out-of-sample level term factors for variance swap returns.

Maturity	Predictor	R2_OOS	CW stat	CW p Val	DA	DA p Val	N_OOS
3m	γ_0^{180-30}						
3m	γ_0^{360-30}						
3m	$\gamma_0^{360-180}$						
4m	γ_0^{180-30}	-5.3399	-1.594	0.9445	0.25	0.9375	65
4m	γ_0^{360-30}	-6.1281	-1.58	0.9429	0.25	0.9375	65
4m	$\gamma_0^{360-180}$	-0.3916	-0.739	0.77	1	0.0625	65
6m	γ_0^{180-30}	-0.2119	0.255	0.3995	0.7143	0.2266	145
6m	γ_0^{360-30}	-0.2595	-1.374	0.9152	0.4286	0.7734	145
6m	$\gamma_0^{360-180}$	-0.9014	0.192	0.4237	0.4286	0.7734	145
12m	γ_0^{180-30}	0.006	0.546	0.2927	0.7143	0.0898	283
12m	γ_0^{360-30}	-0.0497	0.485	0.3138	0.7143	0.0898	283
12m	$\gamma_0^{360-180}$	-0.0631	0.776	0.2189	0.7143	0.0898	283

Table presents out-of-sample evaluation metrics for variance swap returns.

7.4. APPENDIX A1: STRADDLE RETURNS.

This appendix derives the delta-neutral straddle returns as in Coval and Shumway. (2001) based on Heston (1993) stochastic volatility model which shows that instantaneous excess straddle returns are proportional to innovations in volatility. Assume that the asset price S_t at time t and its variance V_t follow the diffusions:

$$dS_t = (\mu - \delta)S_t dt + S_t \sqrt{V_t} dZ_{1t}, \quad A1 (1)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma\sqrt{V_t}dZ_{2t}, \quad A1 (2)$$

where Z_{1t} and dZ_{2t} are P-Brownian motions with correlation ρ , δ is the dividend yield and μ represents the instantaneous expected rate of return on the underlying asset.

Finally, the authors derive the expression for delta-neutral straddle returns where STR denotes the price of portfolio:

$$\begin{aligned} \frac{dSTR}{STR} = rdt + \frac{\partial STR}{\partial V} \cdot \frac{1}{STR} dV_t - \frac{\partial STR}{\partial V} \\ \cdot \frac{1}{STR} [\kappa(\theta - V_t) - \lambda(S_t, V_t, t)]dt \end{aligned} \quad A1 (4)$$

Such that,

$$\frac{dSTR}{STR} = rdt + \frac{\partial STR}{\partial V} \cdot \frac{1}{STR} (dV_t - E_t^{\mathbb{Q}}[dV_t]) \quad A1 (5)$$

Where $\mathbb{Q}$ is the equivalent martingale measure which implies that excess straddle returns are locally proportion to innovations in volatility. Therefore, it can act as an appropriate proxy for the instantaneous VRP.